

Analysis group meeting

Mathieu Ouillon
09/24/24



MISSISSIPPI STATE
UNIVERSITY™

Cooking chef

- Produce timelines yesterday with all the 620 runs except 18314.
- Start cooking for FTOF calib :
 - 18309, 18318, 18325, 18335, 18379, 18425, 18437, 18451, 18476, 18494, 18547, 18598, 18614, 18670, 18696, 18712, 18730, 18772, 18813, 18838, 18864, 18880, 18926, 19040, 19068, 19111
 - Using coatjava: 11.0.1, yaml schema: calib, train: calib
 - Progress: 65% of the jobs have been done, but mostly decoding and merging, started to produce recon files ~600 files, no train files produced.
- Skim the data from pass0v4, to keep event with a trigger electron:
 - Cuts electron: status <0, pid=11, -10<chi2pid<10
 - Path: /cache/hallb/scratch/rg-d/production/skim_pass0v4/
 - One hipo file per run: skim_run_018468.hipo

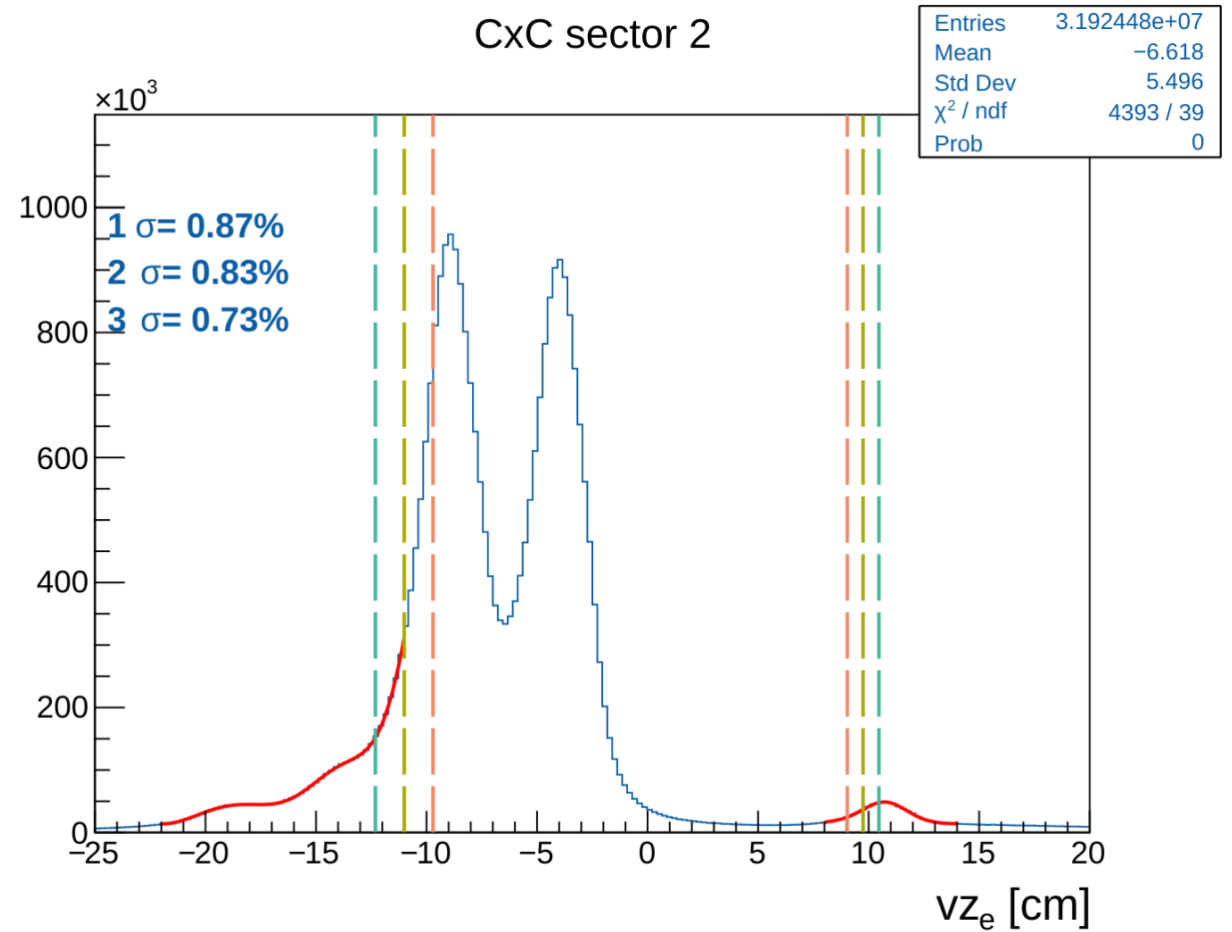
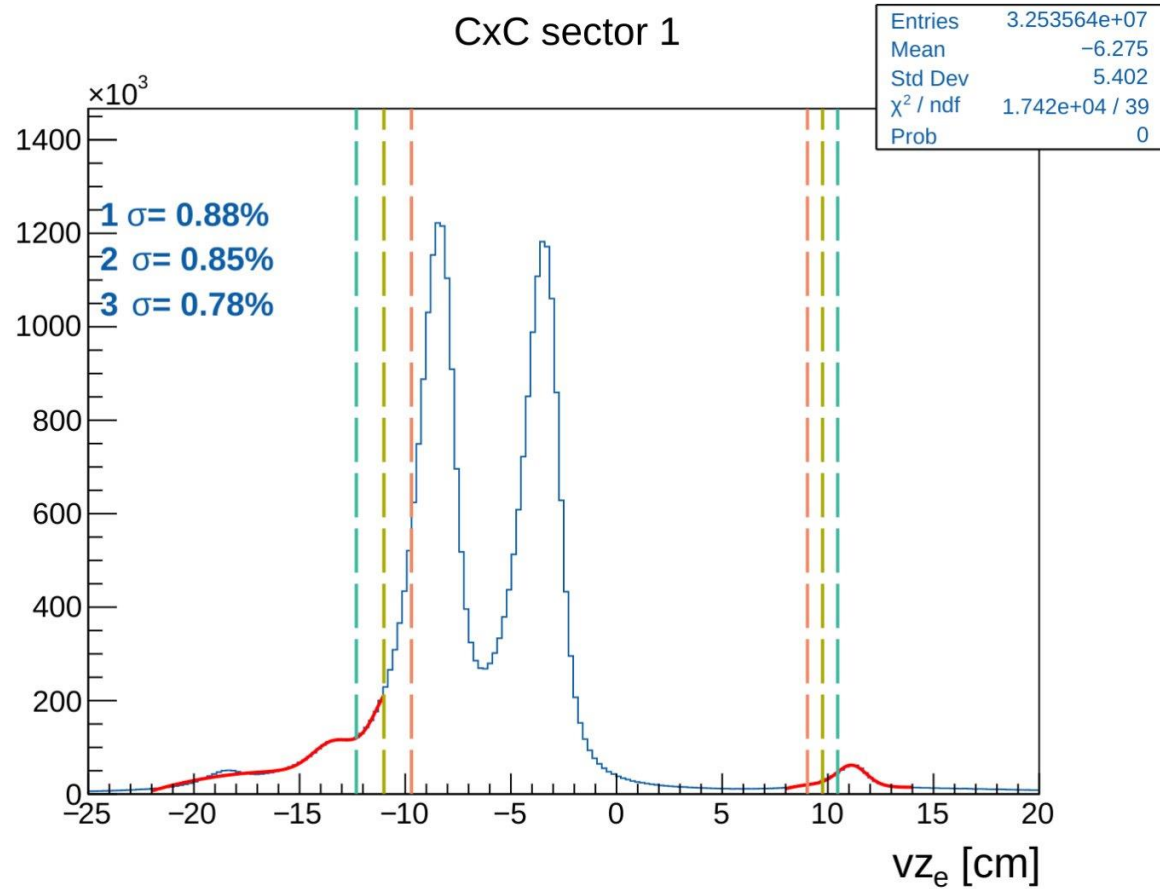
RG-D analysis: vz study

- All Bspot CxC data :
 - /cache/hallb/scratch/rg-d/production/Bspot/v5dstCxC/dst/recon/

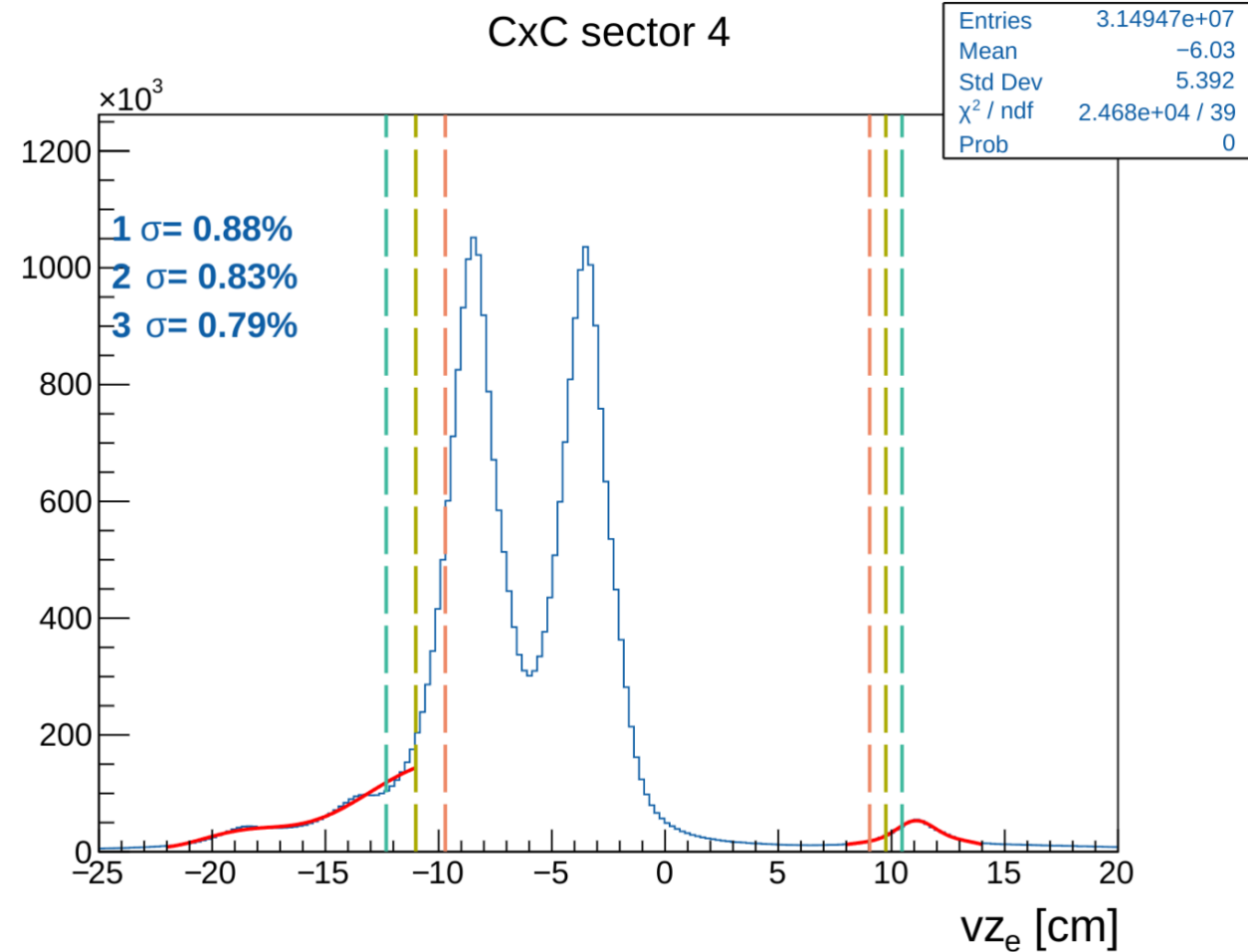
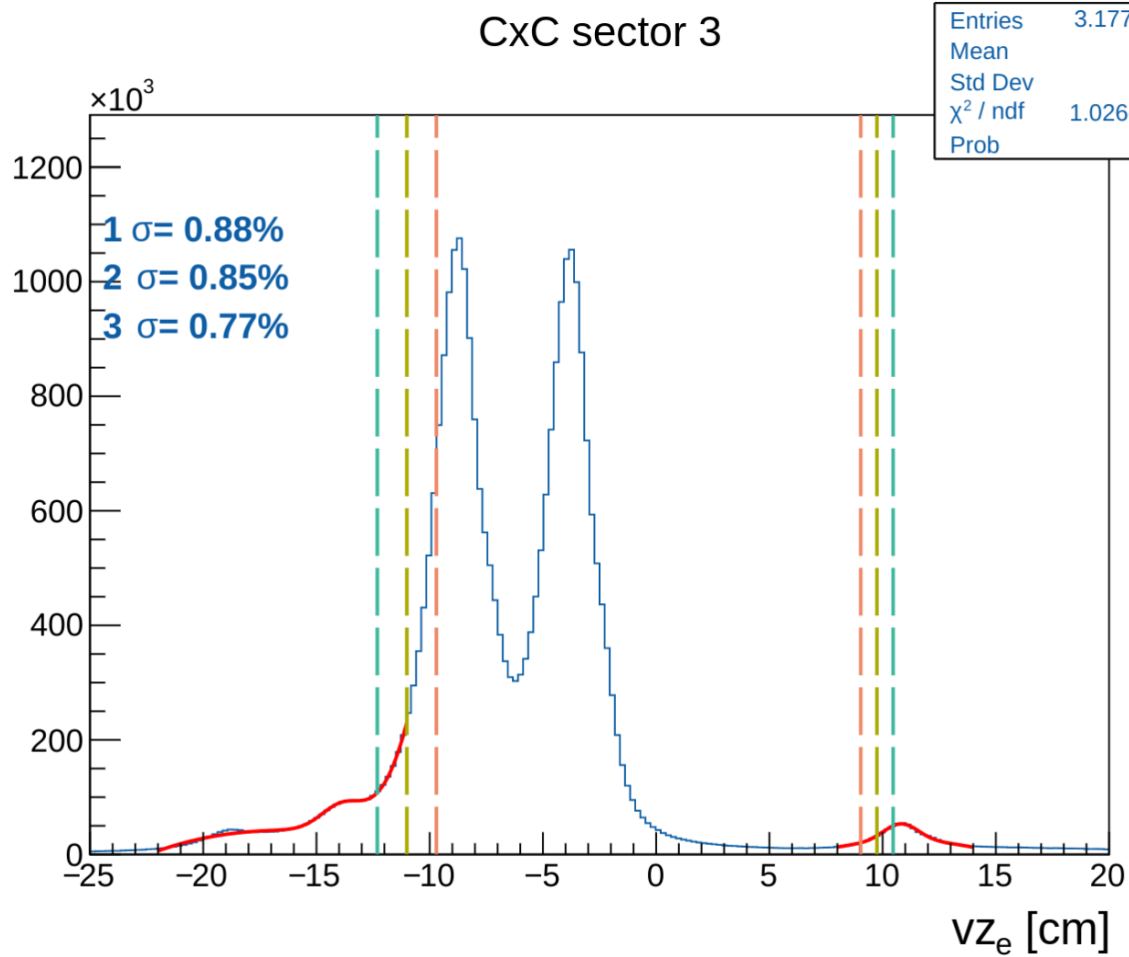
Sector	Mean LD2	Sigma LD2	Mean scat	Sigma scat
1	-13.6028	1.14993	11.1882	0.587501
2	X	X	10.7285	1.00614
3	-13.826	1.30239	10.8061	0.771387
4	-13.4207	1.4602	11.0865	0.766707
5	X	X	11.6962	0.60111
6	X	X	11.5753	0.560613

For the fit: mean LD2 = -13.6165, sigma LD2 = 1.304 , mean scat = 11.1801
Sigma scat = 0.715576

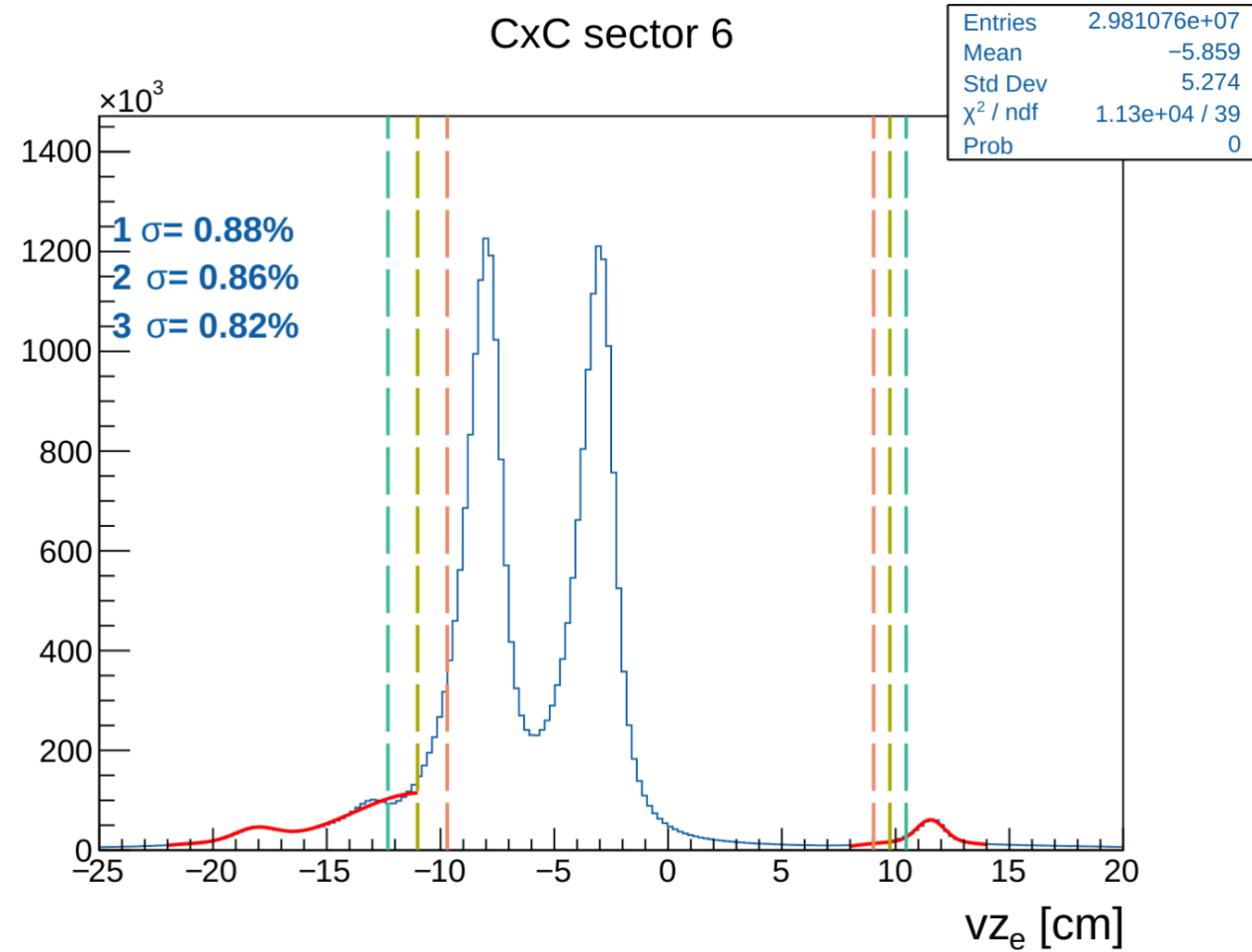
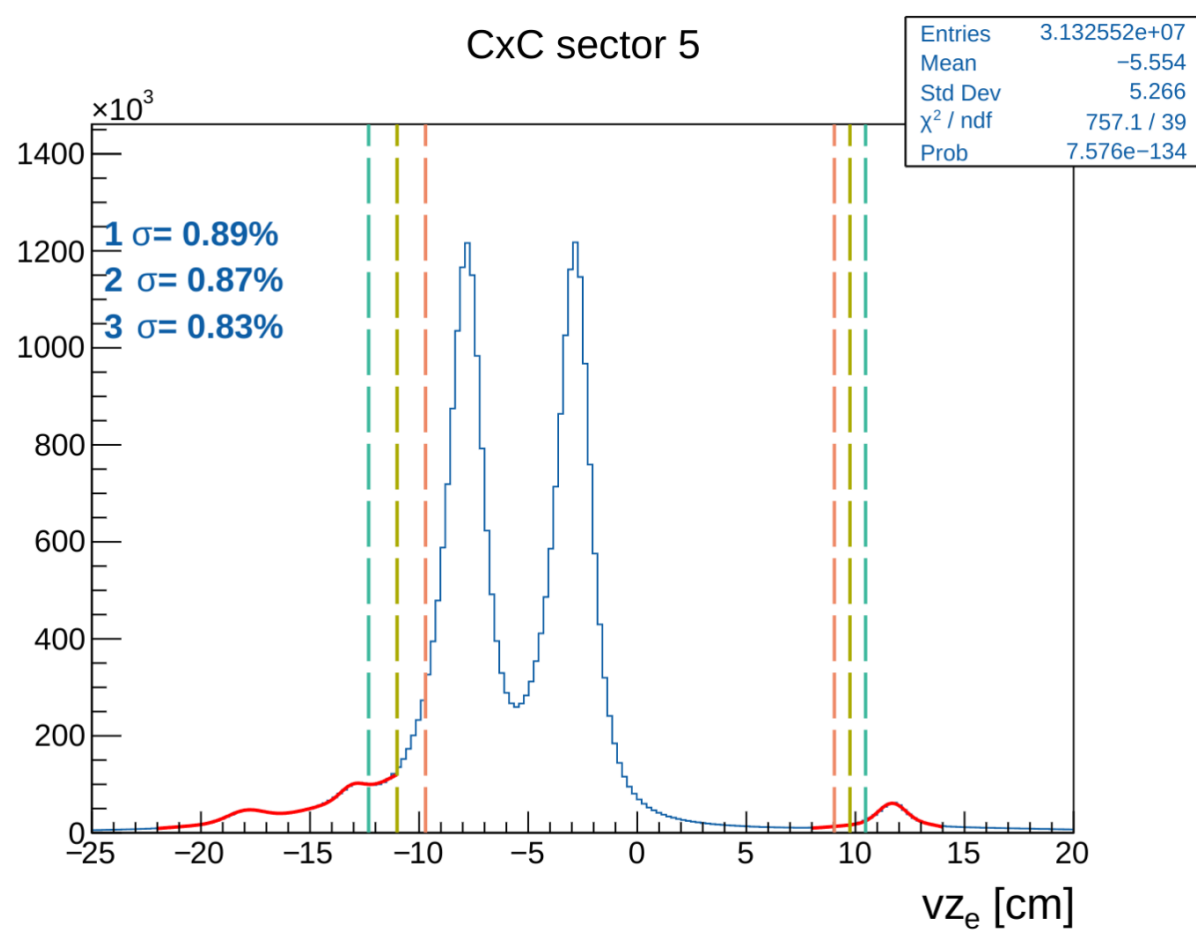
RG-D analysis: vz study



RG-D analysis: vz study



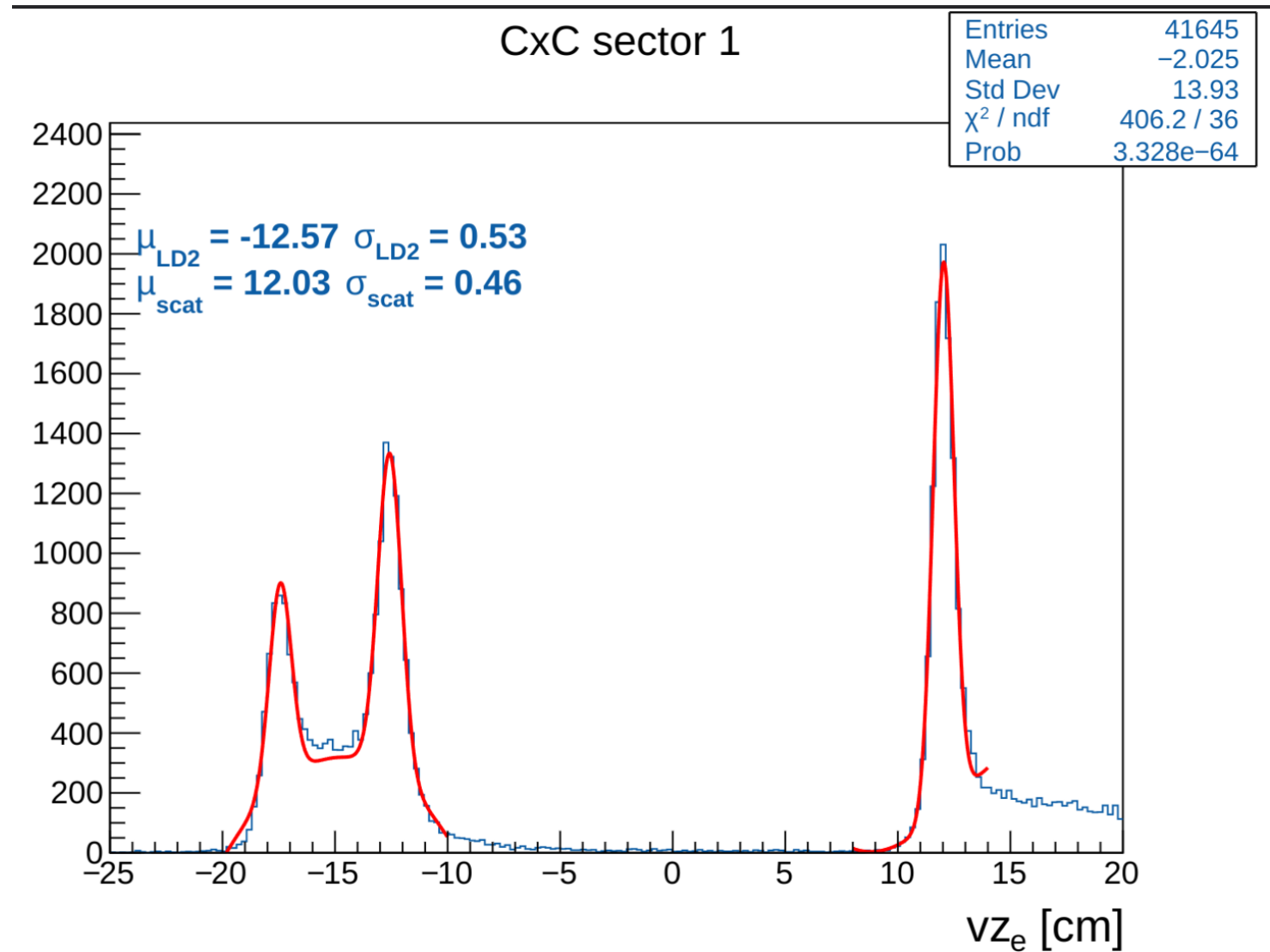
RG-D analysis: vz study



RG-D analysis: vz study

Use pass0v4 data:

- run 018399 empty target



RG-D analysis: Matt analysis crosscheck

Cuts for CxC and LD2 :

Electron :

- Status < 0
- Pid == 11
- $-5 < \chi^2_{pid} < 5$
- $-12 < v_z < 5$ cm

Pion +:

- Pid == 211
- $-10 < \chi^2_{pid} < 10$

Pion -:

- Pid == -211
- $-10 < \chi^2_{pid} < 10$

$W > 2$

$0.1 < -t < 0.5$

$Z_h > 0.9$

$L_c \leq 0.5$

All CxC Bsport runs : 018339, 018340, 018341, 018342, 018343, 018344, 018346, 018440, 018441, 018442, 018443, 018444, 018445, 018475, 018498, 018524, 018756, 018850

- Number of rho0 for each Q2 bins :

- Bin1 = 2278.86 | 2279.495 $\pm$ 86.593 (Matt, mine)
- Bin2 = 1701.1 | 1701.112 $\pm$ 72.793
- Bin3 = 1232.57 | 1232.531 $\pm$ 61.106
- Bin4 = 712.629 | 712.664 $\pm$ 44.349
- Bin5 = 569.64 | 569.618 $\pm$ 42.330
- Bin6 = 215.334 | 217.960 $\pm$ 27.687

- Also check number of LD2 data: same nb of electron, pion+, pion-, and number of rho0 in each bins.

- Bin1 = 6194.99 | 6194.992 $\pm$ 147.981

ALERT: AI for track finding

- Test the model on the hits and doca.
- Bit more complicated, allow tracks to have from 5 to 16 hits.
 - Fill the rest with 0.
- Use a MLP with 48 inputs, 3 hidden layers for 100 neurons (activation function : ReLU), a single output with a sigmoid function.
- Got a accuracy of 92% on the test sample.
- The difficulty come when you want to test all combination of possible tracks from a list of hits :
 - Define all tracks from 5 to 16 hits is a lot.

Confusion Matrix

actual values	False	True
	15753	2657
False	263	17898
True		
predicted values		

Do to list

- Cooking chef:
 - Continue to check the calib workflows.
- RGD analysis:
 - Talk about my analysis.
 - Cross check with Daniel.
- AI ALERT:
 - MLP accuracy if the random hit are 1/2/3/4/... wire away of the track
 - MLP efficiency vs conventional track finding
 - MLP efficiency as a function of the background occupancy
 - Find other architecture do to the track finding faster
- Presentation are available here : <https://github.com/mathieuouillon/Presentation>